ECON 0100 | Exercise C1 | Externalities

Solutions

Answers and work are shown in the red blocks.

Hogsmeade has seen a recent proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented using the following relationships.

$$D : P = 100 - Q_d \qquad\qquad S : P = 10 + Q_s$$

The proliferation of smoked toffee makers has meant the town saw a subsequent rise in *smoke*. Wood smoke can cause lung issues and carries a cost of 10 Galleons to the health of Hogsmeade residents.

Q1 | One Toffee Exchange

First Exchange: A buyer values one smoked toffee at 60 Galleons. It costs the seller 55 Galleons to make.

a) Will the buyer and seller make this exchange? b) Total Surplus to the buyer and seller: c) Marginal Social Cost: d) Total Surplus, including the residents: e) Deadweight Loss from this exchange:

a) Yes. **b)** 5 Galleons. **c)** 65 Galleons. **d)** -5 Galleons. **e)** 5 Galleons.

a) The buyer's value of 60 is above the seller's cost of 55, so there is surplus to split and the trade happens. The smoke lands on the residents, who are not part of the deal, so the buyer and seller have no reason to count it.

b) $60 - 55 = 5$ Galleons for the buyer and seller to share.

c) The cost to society of making this toffee is the seller's cost plus the smoke: $55 + 10 = 65$ Galleons.

d) $60 - 65 = -5$ Galleons. The buyer and seller gain 5, but the residents lose 10.

e) 5 Galleons. The toffee is worth 60 and costs society 65, so making it destroys 5 of surplus. Privately the trade looks good; socially it should not happen.

Another buyer values one smoked toffee at 80 Galleons. It costs the seller 40 Galleons to make.

f) Will the buyer and seller make this exchange? g) Total Surplus to the buyer and seller: h) Marginal Social Cost: i) Total Surplus, including the residents: j) Deadweight Loss from removing this exchange:

> **f)** Yes. **g)** 40 Galleons. **h)** 50 Galleons. **i)** 30 Galleons. **j)** 30 Galleons.
>
> **f)** $80 > 40$, so the trade happens.
>
> **g)** $80 - 40 = 40$ Galleons.
>
> **h)** $40 + 10 = 50$ Galleons.
>
> **i)** $80 - 50 = 30$ Galleons. Even after paying for the smoke, this toffee is worth more than it costs society.
>
> **j)** 30 Galleons. This trade belongs in the efficient market, so banning it (say, banning toffee outright) would throw away the 30 it creates. The goal is not zero toffee but only the toffees worth their full cost.

Q2 | Market Equilibrium

Use a graph and algebra to find equilibrium price, quantity, and DWL. Be sure to show your work.

> Marginal Social Cost: $P = 20 + Q$. Market Equilibrium Price: 55. Market Equilibrium Quantity: 45. Socially Efficient Quantity: 40. Deadweight Loss: 25.

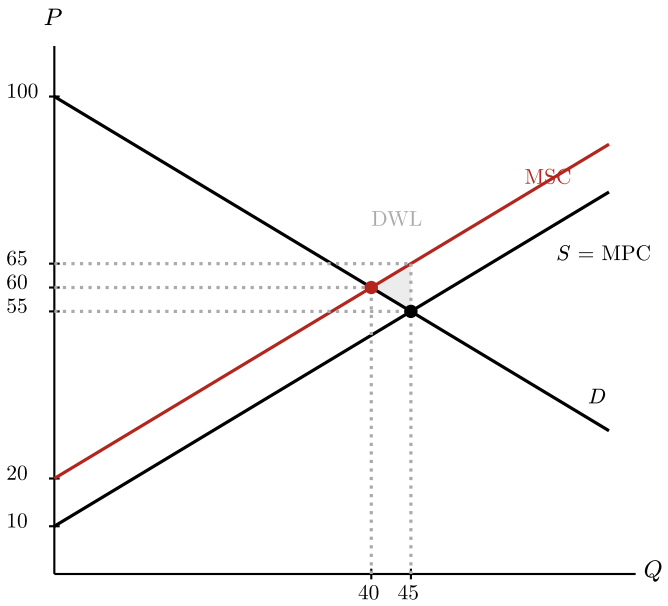


Figure 1: Smoked toffee market with the smoke cost

Every toffee adds 10 of smoke cost on top of the seller's cost, so marginal social cost is supply shifted up by 10:

$$MSC : P = (10 + Q) + 10 = 20 + Q$$

The market only sees the buyers' and sellers' curves, so it settles where demand meets supply:

$$100 - Q = 10 + Q \quad \Rightarrow \quad Q = 45, \quad P = 55$$

The efficient quantity counts the smoke, so it sits where demand meets marginal social cost:

$$100 - Q = 20 + Q \quad \Rightarrow \quad Q = 40, \quad P = 60$$

The market makes 5 toffees too many. Each of the 41st through 45th toffees is worth more to its buyer than it costs its seller, so it trades, but it is worth less than it costs society once the smoke is counted. The 45th, for example, is worth 55 and costs society 65. The losses on those toffees add up to the triangle between marginal social cost and demand from 40 to 45:

$$\text{DWL} = \tfrac{1}{2} \times (65 - 55) \times (45 - 40) = 25$$

Q3 | Yard Flowers

As a small residential town, Hogsmeade has both 1) many shops, and 2) many homes near the town center. Residential homeowners often plant flowers in their front yards, contributing to the positive atmosphere in town. There are numerous garden stores selling supplies for planting and growing many varieties. The supply and demand relationships for a dozen (generic) flowers is given by the following equations, with prices in Galleons.

$$D : P = 100 - \frac{1}{2}Q_d \qquad S : P = \frac{1}{2}Q_s$$

Recently, a study found a positive impact on the wellbeing of the community from the flowers planted in the front yards of their neighbors. The impact was estimated to carry a value of 10 Galleons per dozen flowers planted. Use a graph and algebra to calculate the size of the market failure by identifying and calculating the deadweight loss.

Market Equilibrium Quantity: 100 dozen. Socially Efficient Quantity: 110 dozen. Deadweight Loss: 50 Galleons.

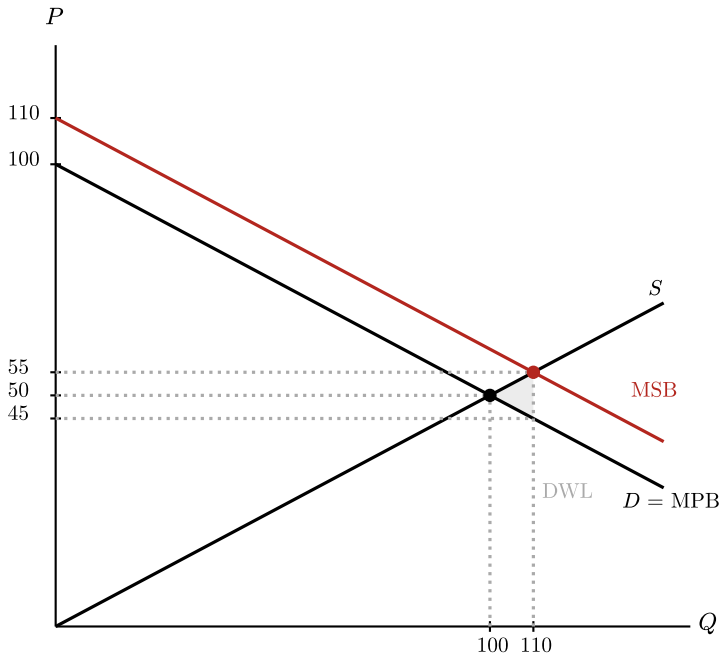


Figure 2: Yard flower market with the neighbors' benefit

Demand is each homeowner's own value of a dozen flowers, the marginal private benefit. The neighbors enjoy another 10 per dozen, so marginal social benefit is demand shifted up by 10:

$$MSB: P = \left(100 - \tfrac{1}{2}Q\right) + 10 = 110 - \tfrac{1}{2}Q$$

The market settles where demand meets supply:

$$100 - \tfrac{1}{2}Q = \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 100, \quad P = 50$$

The efficient quantity counts the neighbors' benefit, so it sits where marginal social benefit meets supply:

$$110 - \tfrac{1}{2}Q = \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 110, \quad P = 55$$

The market plants 10 dozen too few. A homeowner deciding on one more dozen weighs only her own enjoyment against the price; the neighbors' 10 never enters her decision. So the dozens from 100 to 110, which cost more than she values them but less than they are worth to the whole town, go unplanted. The surplus they would have created is the triangle between marginal social benefit and supply from 100 to 110:

$$\text{DWL} = \tfrac{1}{2} \times (55 - 45) \times (110 - 100) = 50$$